

SETH FRIEND

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EDUCATION

University of Texas at Austin

Austin, TX

Integrated B.S./M.S. in Electrical and Computer Engineering

GPA: 3.79

B.S. expected May 2027; M.S. expected May 2028

- Relevant Coursework: Human Centered Robotics, Algorithms, Embedded Systems, Software Design and Implementation II, Honors Linear Systems/Signals, Digital Signal Processing

EXPERIENCES

Systems Engineering Intern - Hardware in Loop Simulation

Phoenix, AZ

Honeywell Aerospace (ValFac)

June – August 2026

- Engineered an automated Python data pipeline to parse SSI files and generate user-ready Excel workbooks, eliminating manual data formatting for Acceptance Test Procedures (ATP) documentation
- Integrated new signals into a Cessna Citation Sovereign simulation bench, completing wiring, software configuration, and end-to-end testing.
- Executed ATP for a Bombardier hardware-in-the-loop simulation bench, supporting validation of real avionics hardware

Electromechanical Lead

Austin, TX

Ivy Technology / Amazon Robotics

May – July 2025

- Led and coordinated teams of 7–10 to build 12 Cardinal automated work cells, delivering a week ahead of schedule
- Installed and wired 400+ automation-ready chutes in collaboration with other leads, integrating mechanical and electrical systems.
- Conducted quality assurance and troubleshooting on 20 Linux-based self-driving robots

Technical Assistant and Design Lead

Austin, TX

Texas Inventionworks | Design Team

Aug 2024 – Present

- Lead and support multidisciplinary student projects, providing guidance on embedded systems, fabrication, and design workflows

Machine Learning Researcher and Robotics Mentor

Austin, TX

Robot Interactive Intelligence Lab

Jan – Dec 2024

- Achieved 95% success training a robot arm in simulation using imitation learning (robomimic)
- Collaborated on a team of 3 to integrate a motion capture system as an input device for Teleoma
- Configured Ubuntu/ROS environments for lab computers, supporting ongoing robotics projects
- Provided individualized help to 20+ high school students in PyTorch, ROS, and kinematics

PROJECTS

Humanoid Robot | *Mechatronics, Controls, Reinforcement Learning*

July 2025 – Present

- Built a 7 kg, 20-DoF humanoid robot with an approximately \$1,700 bill of materials; led lower-body mechanics, torso design, electronics architecture, and software
- Developed Python tools for actuator initialization, encoder validation, and motor testing over CAN bus
- Built an Isaac Lab simulation environment and trained a reinforcement learning policy for walking under external disturbances
- Demonstrated standing on physical hardware; working toward transfer of learned locomotion policies from simulation

EXTRACURRICULAR EXPERIENCES

Engineers for a Sustainable World | *Active Member*

Aug 2023 – Present

Society of Hispanic Professional Engineers | *Active Member*

Aug 2023 – Present

TECHNICAL SKILLS

Languages: Python, C++, C, Java

Software Tools: Git, Linux, VS Code, Bash, Excel/OpenPyX

Libraries: Pandas, NumPy, Matplotlib, PyTorch